STAT-271: Descriptive Statistics and Data Analysis Study Guide

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This study guide covers fundamental concepts in descriptive and inferential statistics based on STAT-271 course notes. It introduces methods of data collection, graphical and tabular data representations, numerical summaries for center and spread, relative positioning with percentiles and quartiles, data distribution shapes, and linear correlation between paired variables.

Introduction to Statistics

- Statistics: The art of learning from data, used to evaluate observations and make informed decisions.
- Observation examples: Average temperatures, unemployment rates, inflation rates, game goals, work productivity, happiness indices, and student test scores.
- Descriptive statistics: Procedures used to describe, organize, summarize, and analyze data using numerical summaries and graphs.
- Inferential statistics: Procedures used to interpret information, generalize sample or experimental results to a larger group, and draw conclusions or make decisions.

Data Collection Methods

- Observational data: Information gathered without intervention by the researcher, such as surveys, where variables are measured without attempting to influence responses.
- Experimental data: Information collected from a designed experiment where a treatment is actively imposed to measure its effect on a response variable.
- Independent variable: The variable manipulated by the researcher (for example, different teaching methods).
- Dependent variable: The response or outcome measured (for example, student test scores).
- Random selection: An essential requirement where all potential group members have an equal likelihood of selection, ensuring valid conclusions.

Statistical Inference, Populations, and Samples

- Probability model: The totality of assumptions made regarding the probabilities of observing different data values.
- Population: The entire set of all individuals or items of interest.
- Sample: A subgroup of the population that is measured and analyzed to draw conclusions about the whole population.
- Representative sample: A sample that accurately reflects the population; typically achieved only through random selection.

- Sampling bias: A systematic distortion favoring certain outcomes, which occurs when a sample is chosen using non-random procedures (such as surveying only attendees at a gym to infer the age of an entire city).

Tabular and Graphical Displays of Data

- Frequency table: Lists distinct data values along with their count of occurrences; best suited for data with relatively few distinct values.
- Relative frequency table: Displays the proportion of occurrences for each value, calculated as frequency divided by total sample size (n).
- Line graph: A plot showing distinct values along the x-axis and vertical line segments whose heights represent frequencies.
- Bar graph: Uses discrete rectangles instead of lines to show frequencies of distinct values.
- Frequency polygon: A line chart formed by directly connecting the top points of the frequency bars or lines.
- Pie chart: Used for non-numerical categorical data, where the area of each circular slice is proportional to its relative frequency.

Grouped Data and Distribution Plots

- Grouped data: Useful when data contains many distinct values; data points are grouped into class intervals (commonly 5 to 10 intervals).
- Interval trade-offs: Too few intervals loses critical information, while too many produces low frequencies and obscures patterns.
- Left-end inclusion convention: The lower boundary is included in the interval, but the upper boundary is excluded.
- Histogram: A display of rectangles placed adjacent to one another with no gaps, representing class frequencies or relative frequencies.
- Ogive: A cumulative frequency or cumulative relative frequency plot that shows the proportion of data points less than or equal to values on the x-axis.
- Stem-and-leaf plot: A compact display for small to moderate datasets where each value is separated into a leading stem and a trailing leaf.

Measures of Central Tendency

- Sample mean: The arithmetic average of all values, calculated as x_bar = (1 / n) * sum(x_i).
- Grouped sample mean: Calculated from a frequency table as x_bar = (1 / n) * sum(f_i * v_i), where v_i are distinct values and f_i are frequencies.
- Linear transformation of the mean: If y_i = a * x_i + b, then y_bar = a * x_bar + b.
- Sample median: The middle value of an ordered dataset. For an odd sample size n, it is the value at position (n + 1) / 2. For an even sample size n, it is 0.5 * (x_(n / 2) + x_((n / 2) + 1)).
- Sample mode: The value that appears most frequently in a dataset; datasets with multiple peaks have multiple modal values.
- Sensitivity to outliers: The mean is strongly affected by extreme values, whereas the median is resistant to extreme values.

Measures of Dispersion

- Sample variance (s^2): The average squared distance from the mean, calculated as s^2 = (1 / (n - 1)) * sum((x_i - x_bar)^2).
- Alternative variance identity: sum((x_i - x_bar)^2) = sum(x_i^2) - n * (x_bar)^2.
- Sample standard deviation (s): The positive square root of sample variance, measuring data spread in the original measurement units.
- Units of measurement: Standard deviation shares the same units as the data, while variance is expressed in squared units.

Percentiles, Quartiles, and Box Plots

- Sample 100p percentile: A value such that at least 100p percent of values are less than or equal to it, and at least 100(1 - p) percent are greater than or equal to it.
- Percentile formula: If n * p is not an integer, take the ordered value at index [n * p + 1], where brackets denote the integer part. If n * p is an integer, take 0.5 * (x_(n * p) + x_(n * p + 1)).
- Quartiles: The 1st quartile corresponds to the 25th percentile, the 2nd quartile is the 50th percentile (median), and the 3rd quartile is the 75th percentile.
- Box plot: A visual five-number summary showing the minimum, 1st quartile, median, 3rd quartile, and maximum.

Distribution Shapes

- Normal datasets: Large datasets with bell-shaped, symmetric histograms where the sample mean, median, and mode coincide at the center peak.
- Negatively skewed (skewed to the left): Distributions with an extended tail toward lower values.
- Positively skewed (skewed to the right): Distributions with an extended tail toward higher values.

Paired Datasets and Correlation

- Scatterplot: A two-dimensional graph showing paired observations (x_i, y_i) to visualize their relationship.
- Sample correlation coefficient (r): Measures the strength and direction of the linear relationship, calculated as r = sum((x_i - x_bar) * (y_i - y_bar)) / ((n - 1) * s_x * s_y).
- Range of r: -1 <= r <= 1.
- Unitless: The correlation coefficient r has no units of measurement.
- Linear transformations: For transformed variables a + b * x and c + d * y, the new correlation is sign(b * d) * r_(x, y).
- Perfect linear relation: If y_i = a + b * x_i, then r = 1 when b > 0, and r = -1 when b < 0.
- Linear limitation: Correlation only evaluates linear association; non-linear relationships can have r = 0 even when variables are strongly dependent.
- Association versus causation: A strong correlation establishes statistical association, but it does not prove that one variable causes changes in the other.